

WAKING STATE REFLEX (Real-Time Read/Write — Sub-25ms Budget)

Dialogue Turn

Raw Utterances

Triage & Embed

MiniLM Encoder

Redis Streams

Async DLQ Ingest

Dual-Brain Stores

Redis + Neo4j GDS

User Query

Dynamic Intent

Quad-Leg Retrieval

HNSW + BM25 + PPR

RRF Aggregator

Rank Fusion

Downstream Prompt

Local Gemma-4



Decoupled Asynchronous Consolidation

DREAMING STATE (Offline Synaptic Consolidation — Background Worker Every 6h)

1. HNSW-DBSCAN

$O(N \log N)$ Dedup
No $O(N^2)$ memory

2. Louvain Partition

Thematic Modules
Centroid Detection

3. God Node Centrality

Stationary PageRank
Epistemic Macro-Hubs

4. CATD Synaptic Pruning

Load-Bearing Weight
 $N_{\text{grace}} \geq 4$ bound